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ECE 1
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Signals lab6

```
fs_orig = 44100;
t_orig = 0 : 1/fs_orig : 1-(1/fs_orig);
L_orig = length(t_orig);

x_orig = cos(2*pi*2000*t_orig) + 0.5*sin(2*pi*4000*(t_orig - pi/4));

X_orig = fft(x_orig);
f_orig = (0 : L_orig-1) * (fs_orig / L_orig);

X_orig_shifted = fftshift(X_orig);
f_orig_shifted = (-L_orig/2 : L_orig/2 - 1) * (fs_orig / L_orig);

fs_target = 48000;
[p, q] = rat(fs_target / fs_orig);

x_resamp = resample(x_orig, p, q);
L_resamp = length(x_resamp);
t_resamp = (0 : L_resamp-1) / fs_target;

X_resamp = fft(x_resamp);
f_resamp = (0 : L_resamp-1) * (fs_target / L_resamp);

X_resamp_shifted = fftshift(X_resamp);
f_resamp_shifted = (-L_resamp/2 : L_resamp/2 - 1) * (fs_target / L_resamp);

figure('Name', 'Assignment 6 - Refactored');
t = tiledlayout(3, 2, 'TileSpacing', 'compact', 'Padding', 'compact');
```

```
nexttile;
plot(f_orig, abs(X_orig));
title('Original FFT');

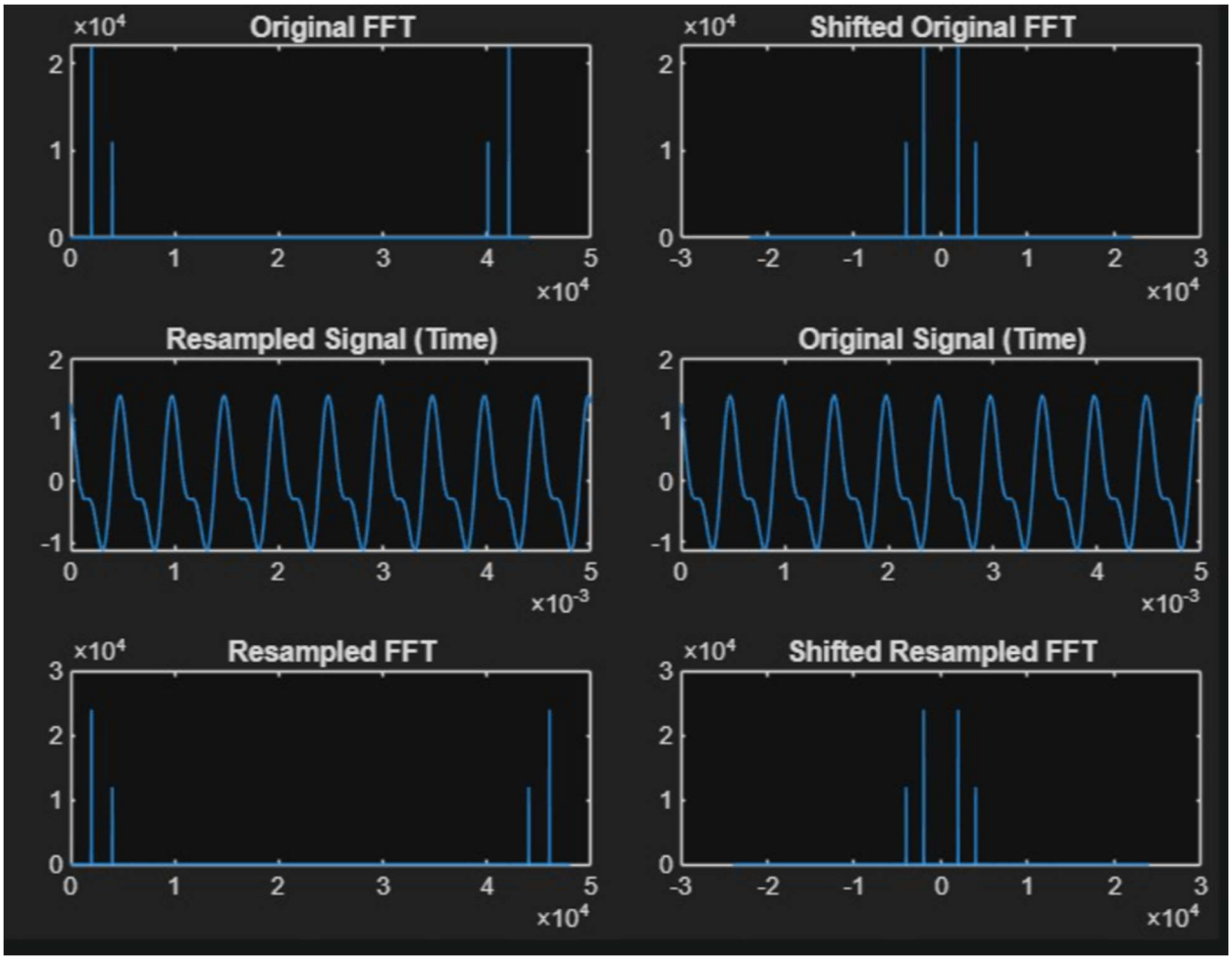
nexttile;
plot(f_orig_shifted, abs(X_orig_shifted));
title('Shifted Original FFT');

nexttile;
plot(t_resamp, x_resamp);
xlim([0 0.005]);
title('Resampled Signal (Time)');

nexttile;
plot(t_orig, x_orig);
xlim([0 0.005]);
title('Original Signal (Time)');

nexttile;
plot(f_resamp, abs(X_resamp));
title('Resampled FFT');

nexttile;
plot(f_resamp_shifted, abs(X_resamp_shifted));
title('Shifted Resampled FFT');
```


Original FFT
×10^4
Shifted Original FFT
×10^4
Resampled Signal (Time)
×10^-3
Original Signal (Time)
×10^-3
Resampled FFT
×10^4
Shifted Resampled FFT
×10^4